

Bocconi

DECISION ANALYSIS



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AGENDA

1. WHY WE NEED A SCIENTIFIC APPROACH TO DECISION-MAKING
2. BASICS ON PROBABILITIES AND DISTRIBUTION
3. MEAN AND STANDARD DEVIATION
4. CONDITIONAL PROBABILITY AND JOINT PROBABILITY
5. PREDICTING OF EVENTS
6. MAKING DECISIONS
7. APPLICATION: SOUTHWEST AIRLINES



1.0

WHY WE NEED A SCIENTIFIC APPROACH TO DECISION-MAKING



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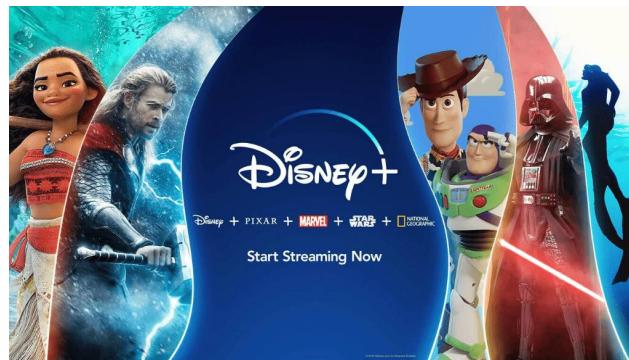
"In just a few short weeks on the job, I had already realized that because every tough decision came down to a probability, then certainty was an impossibility — which could leave me encumbered by the sense that I could never get it quite right. So rather than let myself get paralyzed in the quest for a perfect solution, or succumb to the temptation to just go with my gut every time, I created a sound decision-making process — one where I really listened to the experts, followed the facts, considered my goals and weighed all of that against my principles. Then, no matter how things turned out, I would at least know I had done my level best with the information in front of me."

BARACK OBAMA

Poll time!

How many subscribers did Disney Plus secure in its first year of existence?

- A) 10 millions
- B) 20 millions
- C) 40 millions
- D) 70 millions



In November 2019, Disney launched its new **Disney+** streaming service.

When journalists asked Reed Hastings (the boss of Netflix) how he thought Disney+ was going to perform in its first year, **Hastings said it would secure 20 million subscribers “at best”**.

At the time, Netflix had about 160 million subscribers. Its streaming service was launched in 2007.



Poll time!

How many total ChotuKool units did Godrej sell in the first two years?

- A) 15,000
- B) 300,000
- C) 2,500,000
- D) 12,000,000



- Designed for people of rural areas in India.
- Priced at \$69 (world's cheapest fridge)
- No compressor, Only Cooling chip and a fan run on batteries
- 20 Parts
- Involve villagers right from the design to selling of the product.



In 2008, Indian household appliances manufacturer Godrej introduced **ChotuKool**, a low-cost refrigerator targeted at the hundreds of millions of Indians who could not afford a regular refrigerator or did not have stable access to electricity.

The COO (chief operating officer) of Godrej said that they were expecting to sell “**probably millions**” of units in the first three years.

NOKIA



Ring

Shortcut
Everything



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February 2019

BMW and Daimler announced that **they would invest \$1 billion in Free Now**, their car-sharing joint venture.

Free Now included Share Now, the division centered on short-term car rentals, that already had 4m customers in more than 30 cities under the BMW brand DriveNow and Daimler's Car2go.

The plan was to extend Free Now's services to **another 90 cities** and **grow tenfold** in the next three years.



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BMW-Daimler ride hailing unit plans 'tenfold' expansion

German rivals joint venture Free Now to drive into another 90 cities this year



Harald Krueger, chief executive officer of BMW (left) and Dieter Zetsche, his counterpart at Daimler in Berlin at the launch of their expanded joint venture © Bloomberg

Patrick McGee FEBRUARY 24 2019



FT
FINANCIAL
TIMES

December 2019

BMW and Daimler announced that **Free Now** would leave the United States and Canada due the “volatile state of the global mobility landscape” and the “rising infrastructure complexities facing North American transportation”.

The business also said it **would leave Brussels, London and Florence** due to “low adoption rates”.

DriveNow and Car2Go, which had already pulled out of more than a dozen locations, will continue to operate in 18 European cities, including seven in Germany.



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BMW, Daimler reverse out of car-sharing venture

German carmakers blame rising costs and lack of customer interest



Joe Miller in Frankfurt and Patrick McGee in San Francisco DECEMBER 18 2019



BMW and Daimler are pulling their flagship car-sharing services out of North America and the UK, citing rising costs and insufficient customer interest, as the companies' chief executives refocus on selling premium vehicles.

FT
FINANCIAL
TIMES

German automakers do U-turn on car-sharing push

Fading spectre of ‘peak car’ diminishes appeal of investing in vehicle-sharing and ride-hailing apps

Joe Miller in Frankfurt DECEMBER 24 2019

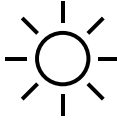
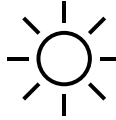


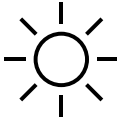
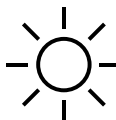
The volte face is indicative of a “welcome outbreak of sanity” in the industry over the past few months, said Bernstein auto analyst Max Warburton. “Many of the so-called ‘legacy’ manufacturers — urged on by consultants, media and yes, equity analysts — got sucked into spending on this stuff.”



Uncertainty rules also in our daily lifes

Think about weather forecast: which one is more informative?

MON	TUE	WED	THU	FRI
				

MON	TUE	WED	THU	FRI
				
Chance of rainfall: 20%	Chance of rainfall: 40%	Chance of rainfall: 90%	Chance of rainfall: 70%	Chance of rainfall: 80%

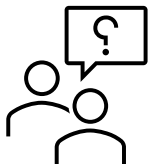


3 main goals

In this class we will learn how to make predictions using probability and updating our beliefs to make better decisions

BE LESS CERTAIN

Overconfidence depends on culture, personality, expertise, ...



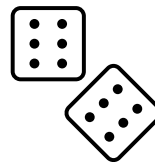
LEARN TO MAKE PREDICTIONS

Question how often an outcome typically happen



THINK PROBABILISTICALLY

Have a theory to envision probability of events



2.0

BASICS ON PROBABILITIES AND DISTRIBUTIONS



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Let's introduce probability

<https://www.khanacademy.org/math/cc-seventh-grade-math/cc-7th-probability-statistics/cc-7th-basic-prob/v/basic-probability>



A random variable is a variable that can take on some values ($x_1, x_2, \dots, x_n$). Each value is assigned a probability ($p_1, p_2, \dots, p_n$)

Random variable	Possible values	Random events	Probability
-----------------	-----------------	---------------	-------------

$X =$

$\left\{ \begin{array}{l} 0 \\ 1 \end{array} \right.$

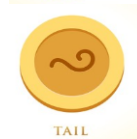
0



HEAD

0.5

1



TAIL

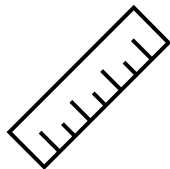
0.5



Random variables - values

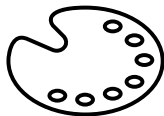
CONTINUOUS

Variables that take on a value in a range



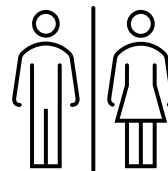
QUALITATIVE / CATEGORICAL

Variables that do not come from measuring or counting



DICHOTOMOUS / BINARY

Variables that can take on only two values



We define **PROBABILITIES**

Probabilities indicate the likelihood of the occurrence of a random event

Probabilities are a sequence of numbers $[0,1]$ that have to sum up to 1



Values	Probabilities	Cumulative Probabilities
10	0.20	0.20
20	0.10	0.30
30	0.20	0.50
40	0.20	0.70
50	0.10	0.80
60	0.05	0.85
70	0.10	0.95
80	0.05	1.00

A random variable X can take on a certain value based on the assigned probability

i.e. the probability of getting a value of $X = 10$ is 0.2 or 20%



Values	Probabilities	Cumulative Probabilities
10	0.125	0.125
20	0.125	0.250
30	0.125	0.375
40	0.125	0.500
50	0.125	0.625
60	0.125	0.750
70	0.125	0.875
80	0.125	1.000

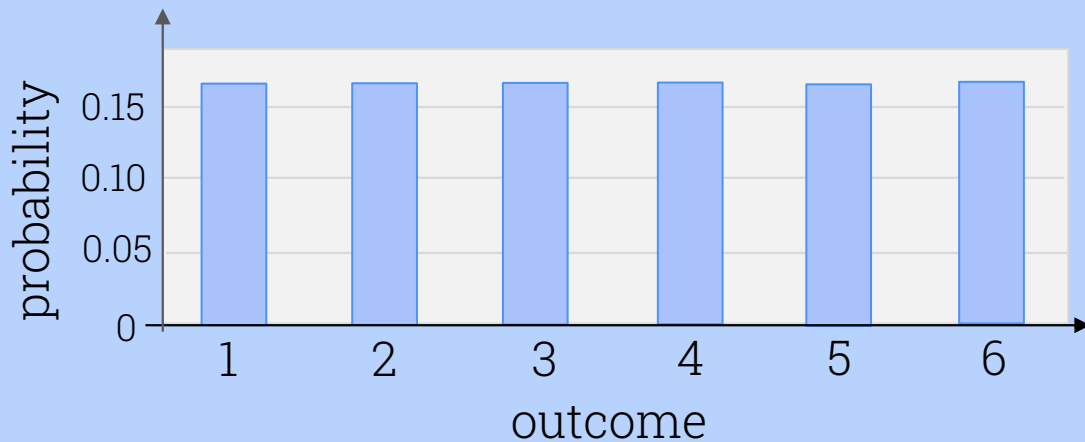
UNIFORM DISTRIBUTION

**Each value has the same probability
X has an equal likelihood of being
any of these 8 outcomes.**

i.e. the variable can take on 8
different values (10, 20, 30, .., 80)
but the probability of each value is
always 12.5%



Graphical representation: rolling a die has a uniform distribution



How informative is the uniform distribution?

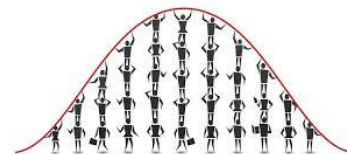


Values	Probabilities	Cumulative Probabilities
10	0.02	0.02
20	0.08	0.10
30	0.15	0.25
40	0.25	0.50
50	0.25	0.75
60	0.15	0.90
70	0.08	0.98
80	0.02	1.00

NORMAL DISTRIBUTION

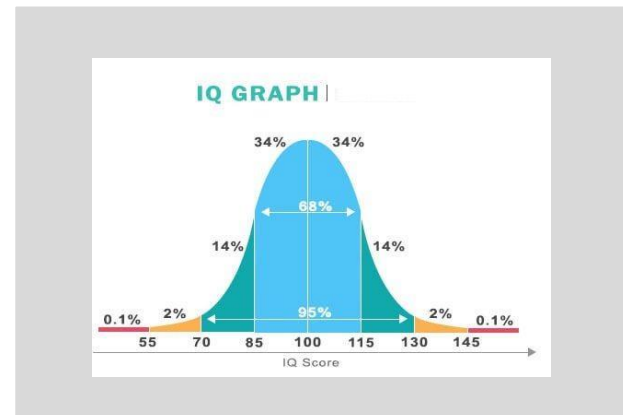
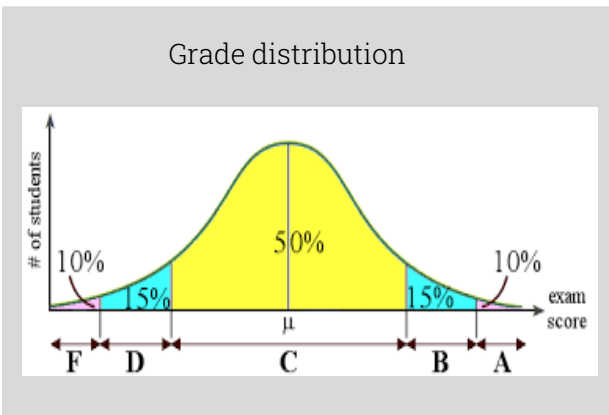
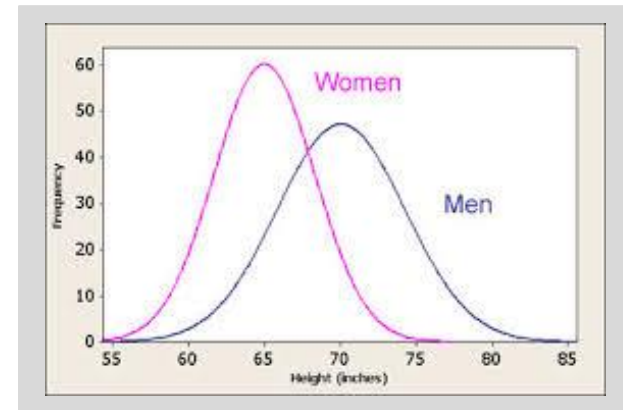
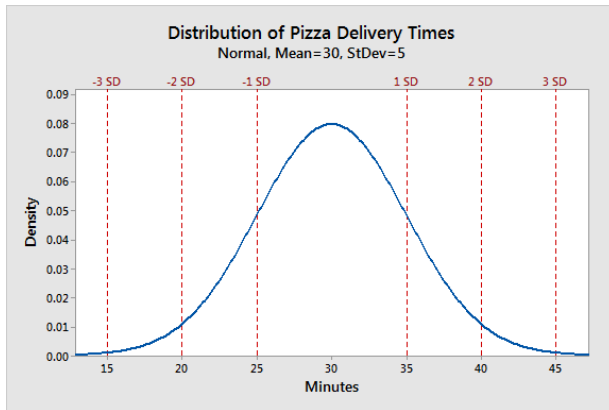
In a normal distribution, the outcomes tend to distribute around a central value

i.e. in the table: Which values are more likely? And which ones are less likely?



Can you think of examples
where you may observe a
normal distribution?

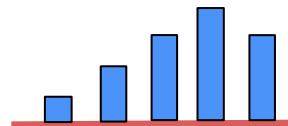




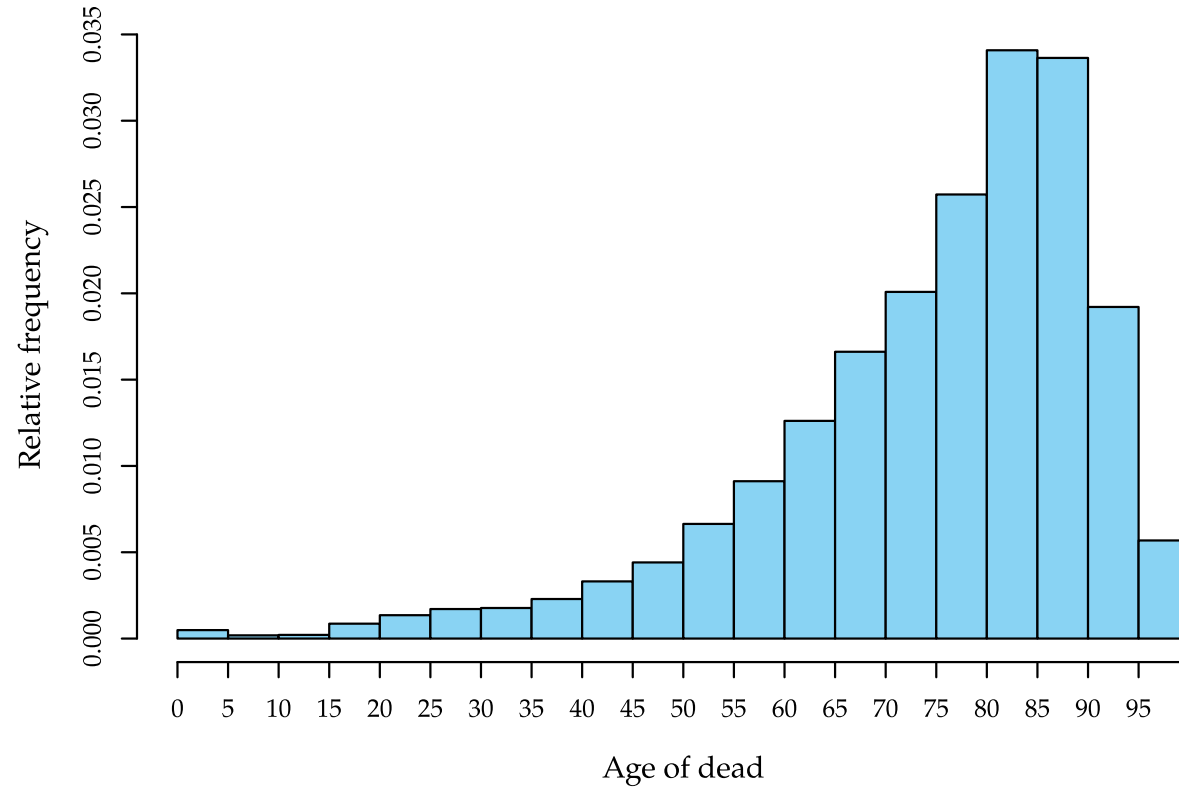
Values	Probabilities	Cumulative Probabilities
10	0.02	0.02
20	0.03	0.05
30	0.05	0.10
40	0.05	0.15
50	0.10	0.25
60	0.15	0.40
70	0.40	0.80
80	0.20	1.00

NORMAL DISTRIBUTION

Left skewed: The values of the observations cluster more around the right side of the distribution



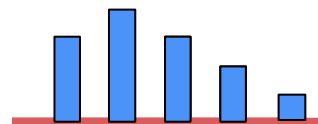
Distribution of age at death of australian males

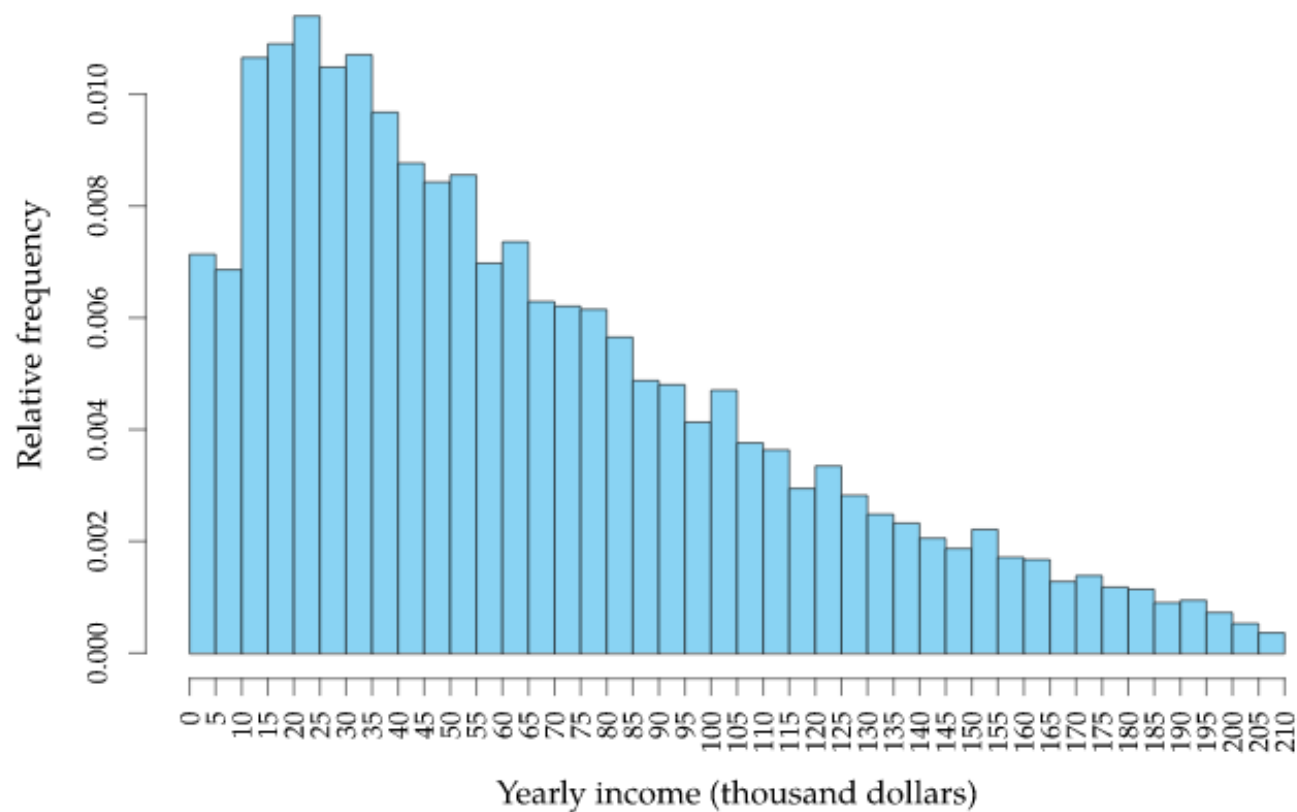


Values	Probabilities	Cumulative Probabilities
10	0.20	0.20
20	0.40	0.60
30	0.15	0.75
40	0.10	0.85
50	0.05	0.90
60	0.05	0.95
70	0.03	0.98
80	0.02	1.00

NORMAL DISTRIBUTION

Right skewed: values cluster more around the left side of the distribution





3.0

MEAN AND STANDARD DEVIATION



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Summary Statistics (or moments)

PROBABILITY
DISTRIBUTIONS HAVE
SUMMARY STATISTICS,
OR MOMENTS

SUMMARY STATISTICS
ARE USED TO
DESCRIBE DATA OR
THE SAMPLE

THE TWO MOST
IMPORTANT
STATISTICS ARE

- Mean
- Standard deviation
(= square root of the
variance)



Do you think Bocconi's
students are tall or short?

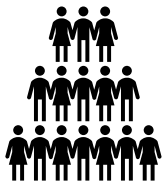


Mean

Probably you have no clue!



Mean = 175 cm



Height of 15,000
students at Bocconi

But if I tell you that the mean of the height of Bocconi students is 175 cm, then what would you answer?

And if now I tell you that the mean of the height of the students at Politecnico is 180 cm, then what would you answer?

**ONE NUMBER CAN TELL YOU A LOT
ABOUT MANY NUMBERS**

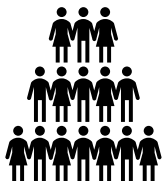


Variance (or standard deviation)

The mean tells you where the center of the distribution is positioned



Mean = 175 cm



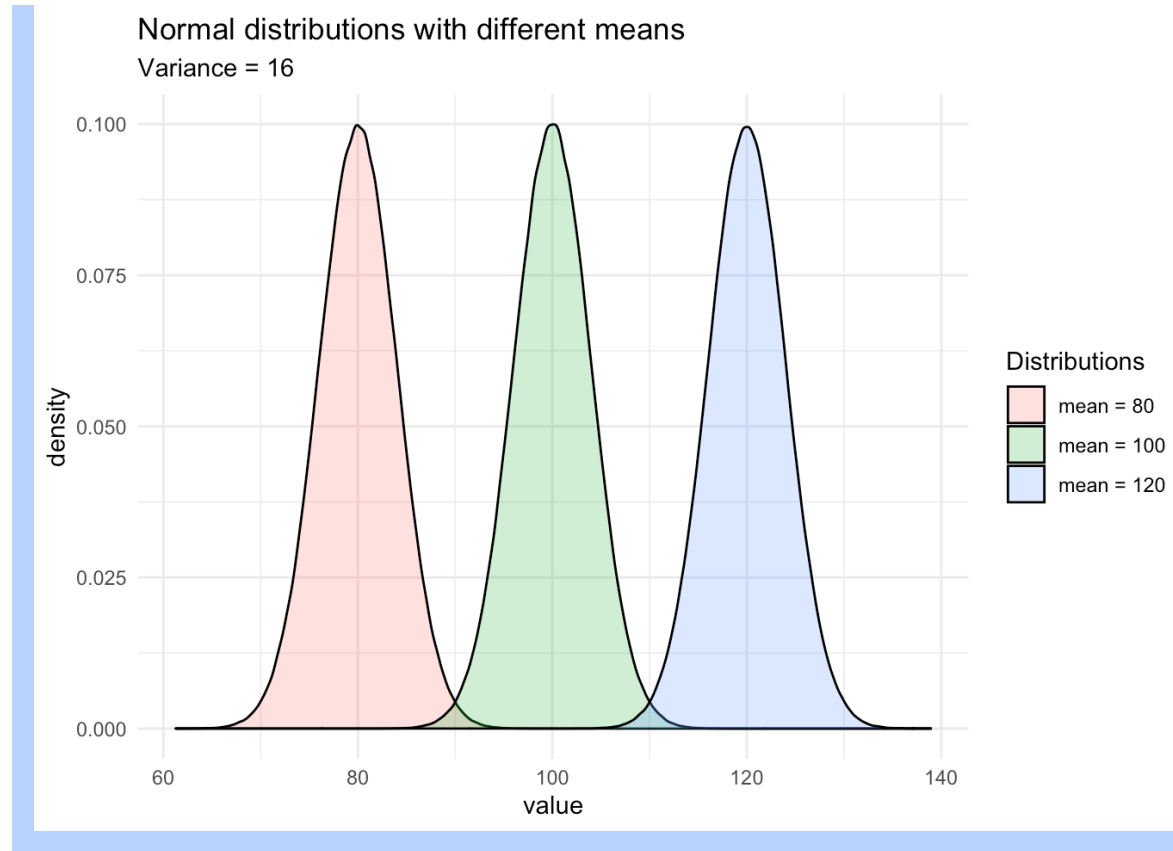
Height of 15,000
students at Bocconi

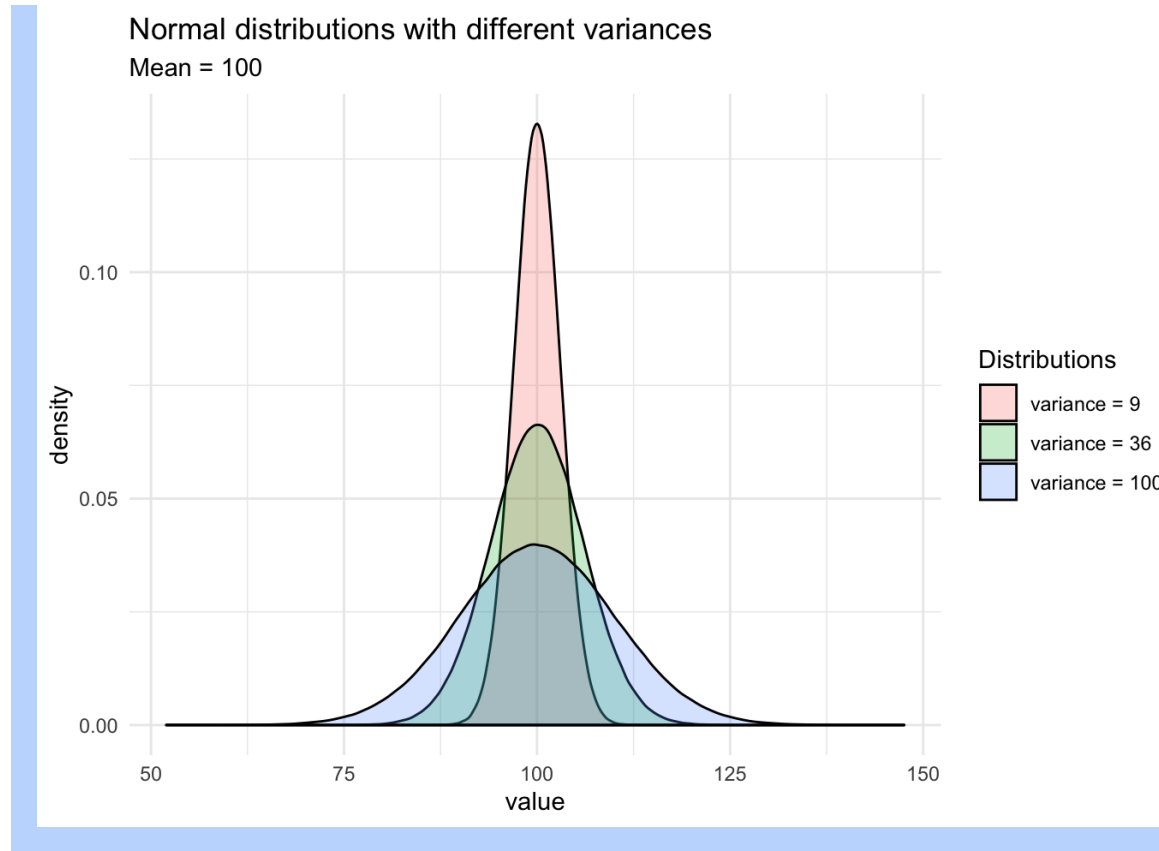
Means do not tell you whether the values of the observations are concentrated around it, or dispersed away from it

Small variance : most of the Bocconi students' heights are around 175 cm

High variance: many Bocconi students are either very short (155 cm) or very tall (195 cm)







Values	Probabilities	Cumulative Probabilities
10	0.20	0.20
20	0.10	0.30
30	0.20	0.50
40	0.20	0.70
50	0.10	0.80
60	0.05	0.85
70	0.10	0.95
80	0.05	1.00

Mean formula: $\mu = \sum_{i=1}^n p_i x_i$,

Where x_i is the value of an observation, and p_i is the probability of whether that value occurs.

The mean of the distribution in the table is:

$$10*0.20 + 20*0.10 + 30*0.20 + 40*0.20 + 50*0.10 + 60*0.05 + 70*0.10 + 80*0.05 = \mathbf{37}$$



Values	Probabilities	Cumulative Probabilities
10	0.20	0.20
20	0.10	0.30
30	0.20	0.50
40	0.20	0.70
50	0.10	0.80
60	0.05	0.85
70	0.10	0.95
80	0.05	1.00

Variance formula: $\sigma^2 = \sum_{i=1}^n p_i (x_i - \mu)^2$,

Where x_i is the value of an observation, and p_i is the probability of whether that value occurs

The variance of the distribution in the table is:

$$\begin{aligned}
 &(10 - 37)^2 * 0.20 + (20 - 37)^2 * 0.10 + \\
 &(30 - 37)^2 * 0.20 + (40 - 37)^2 * 0.20 + \\
 &(50 - 37)^2 * 0.10 + (60 - 37)^2 * 0.05 + \\
 &(70 - 37)^2 * 0.10 + (80 - 37)^2 * 0.05 = \mathbf{431}
 \end{aligned}$$

Standard deviation = $\sqrt{\text{Variance}}$

What's the value of the standard deviation of the distribution?



Poll time!

What type of variable is "Family_firm"?

- A) Qualitative
- B) Continuous
- C) Dichotomous

Year	ID	Family_firm	ROE	ROA	Turnover	Sector
2019	1	1	12.10	10.60	58,155.96	Manufacturing
2019	2	0	9.80	6.74	316,792.54	Fashion
2019	3	0	5.42	3.15	22,631.89	Transport and logistics
2019	4	0	2.44	1.85	136,295.32	Metal products
2019	5	0	6.37	2.66	51,651.85	Transport and logistics
2019	6	1	16.59	5.86	48,288.88	Wholesale
2019	7	1	30.41	7.26	152,355.00	Manufacturing
2019	8	1	9.94	5.22	36,519.24	Fashion
2019	9	0	9.40	4.71	194,447.17	Wholesale
2019	10	0	-12.81	-4.35	338,855.24	Metal products



Poll time!

What type of variable is "Turnover"?

- A) Qualitative
- B) Continuous
- C) Dichotomous

Year	ID	Family_firm	ROE	ROA	Turnover	Sector
2019	1	1	12.10	10.60	58,155.96	Manufacturing
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2019	9	0	9.40	4.71	194,447.17	Wholesale
2019	10	0	-12.81	-4.35	338,855.24	Metal products



Poll time!

What type of variable is "Sector"?

- A) Qualitative
- B) Continuous
- C) Binary

Year	ID	Family_firm	ROE	ROA	Turnover	Sector
2019	1	1	12.10	10.60	58,155.96	Manufacturing
2019	2	0	9.80	6.74	316,792.54	Fashion
2019	3	0	5.42	3.15	22,631.89	Transport and logistics
2019	4	0	2.44	1.85	136,295.32	Metal products
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2019	8	1	9.94	5.22	36,519.24	Fashion
2019	9	0	9.40	4.71	194,447.17	Wholesale
2019	10	0	-12.81	-4.35	338,855.24	Metal products



Poll time!

What type of variable is “ROE”?

- A) Qualitative
- B) Continuous
- C) Dichotomous

Year	ID	Family_firm	ROE	ROA	Turnover	Sector
2019	1	1	12.10	10.60	58,155.96	Manufacturing
2019	2	0	9.80	6.74	316,792.54	Fashion
2019	3	0	5.42	3.15	22,631.89	Transport and logistics
2019	4	0	2.44	1.85	136,295.32	Metal products
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2019	9	0	9.40	4.71	194,447.17	Wholesale
2019	10	0	-12.81	-4.35	338,855.24	Metal products



Poll time!

How do you expect turnover to be distributed?

A) Normally distributed

B) Left skewed

C) Right skewed

Year	ID	Family_firm	ROE	ROA	Turnover	Sector
2019	1	1	12.10	10.60	58,155.96	Manufacturing
2019	2	0	9.80	6.74	316,792.54	Fashion
2019	3	0	5.42	3.15	22,631.89	Transport and logistics
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2019	8	1	9.94	5.22	36,519.24	Fashion
2019	9	0	9.40	4.71	194,447.17	Wholesale
2019	10	0	-12.81	-4.35	338,855.24	Metal products



4.0

CONDITIONAL PROBABILITY AND JOINT PROBABILITY



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What is **conditional probability**?

The probability of an event (situation, characteristic, condition, ...) occurring given that another event has already taken place



Conditional probability is everywhere

- If you want to know the probability that your friend has Covid, it is useful to know whether she is vaccinated!
- If you want to know the probability that Messi goes back to Barcelona FC the next season, it is useful to know whether the team can afford to pay him
- If you want to know the probability that customers will buy your new product, it is useful to know what a sample of them think of the test version.



Should Share Now be in Barcelona?

Madrid is currently the only city in Spain in which Share Now operates.

You are an intern at Share Now and your boss asks you to **assess the business case of entering Barcelona**.

That is, they want to know how **probable** it is that Share Now could turn a profit in that city.

Your boss told you that they want the average subscriber to drive at least 20 hours per year.

What **variables** do you think the probability of turning a profit could depend on?



SHARENOW ✓



Should Share Now be in Barcelona?

Consider **per-capita income**.

Is it more likely that Share Now will turn a profit in the city?

- Maybe yes (high-income people do not like to waste time in the traffic and are ready to pay for rental of small cars)
- Maybe not (high-income people like to own their car)
- Maybe it has no impact (not a variable or different effects that balance each other)

You usually need to **check data** to know whether an event (like profit) depends on something else (like per-capita income).

YES!

NO

MAYBE



Expected Probability in Brief

You have just met a person.

You have to guess how many letters she has in her name.

What would you say?

Would you change your answer if I tell you she is from Korea?



Who is more likely to have short last names - Italians or Koreans?



Area of origin					
		Italy	Asia	Others	MARGINAL
# letters in last name (X)	2	0	2	0	2
	3	1	5	0	6
	4	3	1	0	4
	5	9	2	2	13
	6	8	2	4	14
	7	21	0	5	26
	8	9	0	2	11
	9	7	0	2	9
	10	4	0	0	4
	11	1	0	0	1
	14	0	0	2	2
	23	0	0	1	1
MARGINAL		63	12	18	93

The frequency table to the left reports the number of letters of the last name of 93 students-based on whether originate from Italy, Asia or somewhere else

The number of letters in the last name is our random variable X

- This type of table is called a **“contingency table”**: it shows the values of X contingent on another variable.
- The totals of rows and columns are **marginal frequencies** (that is, the “last” row/column).



Area of origin					
		Italy	Asia	Others	MARGINAL
# letters in last name (X)	2	0	2	0	2
	3	1	5	0	6
	4	3	1	0	4
	5	9	2	2	13
	6	8	2	4	14
	7	21	0	5	26
	8	9	0	2	11
	9	7	0	2	9
	10	4	0	0	4
	11	1	0	0	1
	14	0	0	2	2
	23	0	0	1	1
MARGINAL		63	12	18	93

Conditional probability: $P(X|Y)$

What is the probability of a student with 5 letters in his or her last name?

E.g. $13/93 = 13.9\%$

What is the probability of a student with 5 letters in his or her last name given that the student is Italian?

E.g. $P(X=5 | \text{Italy}) = 9/63 = 14.3\%$

The conditioning event is often called a condition, clue, data point or signal (being Italian)



What is **Conditional Expectation**?

The expected value of a random variable, given that another event has already taken place



Area of origin					
		Italy	Asia	Others	MARGINAL
# letters in last name (X)	2	0	2	0	2
	3	1	5	0	6
	4	3	1	0	4
	5	9	2	2	13
	6	8	2	4	14
	7	21	0	5	26
	8	9	0	2	11
	9	7	0	2	9
	10	4	0	0	4
	11	1	0	0	1
	14	0	0	2	2
	23	0	0	1	1
	MARGINAL	63	12	18	93

Conditional expectation:

$$E(X|Y) = \sum_{i=1}^n p(x_i|Y) x_i$$

That is, you multiply all the values by their respective conditional probabilities, contingent on Y, and then you add them up.

$$E(X | \text{Italy}) = 2 * 0/63 + 3 * 1/63 + 4 * 3/63 + 5 * 9/63 + 6 * 8/63 + 7 * 21/63 + 8 * 9/63 + 9 * 7/63 + 10 * 4/63 + 11 * 1/63 + 14 * 0/63 + 23 * 0/63 = 7$$

$E(X | \text{Italy}) = 7 = \#$ of letters the last name will have on average given the student is Italian.



What is **Joint probability**?

The probability that two events take place together at the same time



Area of origin					
		Italy	Asia	Others	MARGINAL
# letters in last name (X)	2	0	2	0	2
	3	1	5	0	6
	4	3	1	0	4
	5	9	2	2	13
	6	8	2	4	14
	7	21	0	5	26
	8	9	0	2	11
	9	7	0	2	9
	10	4	0	0	4
	11	1	0	0	1
	14	0	0	2	2
	23	0	0	1	1
MARGINAL		63	12	18	93

Joint probability:

$$P(X,Y) = P(X|Y)*P(Y)$$

That is, the probability that **Y happens** times the probability that **X happens once we know that Y happened**

E.g.

$$P(X=5, \text{Italy}) = P(X=5 | \text{Italy}) * P(\text{Italy}) = \\ = 9/63 * 63/93 = 14.3\% * 67.7\% = \mathbf{9.7\%} =$$

probability that the student is Italian **AND** the last name has 5 words



Notice the difference between conditional and joint probabilities

CONDITIONAL PROBABILITY

$$P(X|Y)$$

$$P(X=5 | \text{Italy}) = 9/63 = 14.3\%$$

JOINT PROBABILITY

$$P(X,Y) = P(X|Y) * P(Y)$$

$$P(X=5, \text{Italy}) = 9/63 * 63/93 = 9.7\%$$



Let's practice

A TLC company is investigating its retention of customers. It selected a sample of 2,185 customers who signed the contract five years ago. Some of them remained (they still have the contract), others left to other providers.

The company believes that the **age group** of customers could have an impact on the retention rate.

	Gen Z (up to 24)	Millennials (25-40)	Gen X (41-56)	Boomers and older (57+)	TOTAL
Remained	200	115	180	210	705
Left	120	160	530	670	1,480
TOTAL	320	275	710	880	2,185



Let's practice

In this sample:

- what is the probability of **retaining a customer**?
- what is the probability of **retaining a Gen Z customer**?
- what is the probability of **being a Gen Z customer who remains**?

	Gen Z (up to 24)	Millennials (25-40)	Gen X (41-56)	Boomers and older (57+)	TOTAL
Remained	200	115	180	210	705
Left	120	160	530	670	1,480
TOTAL	320	275	710	880	2,185



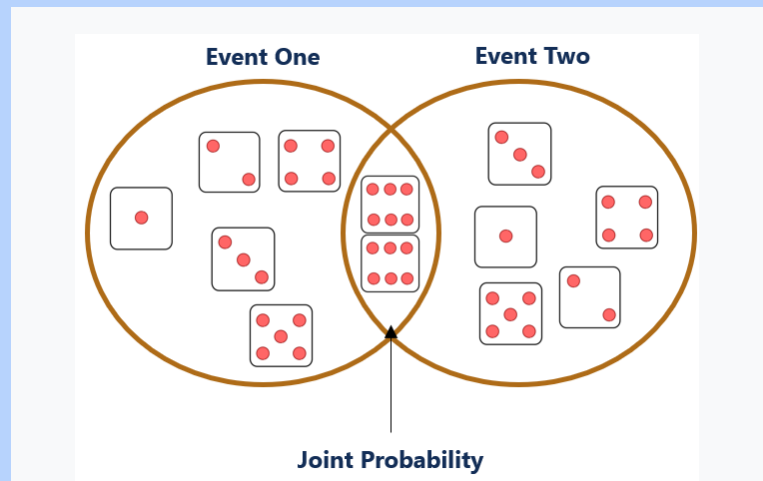
Independence

Do you remember **per-capita income** and Share Now in Barcelona? It is always possible that a variable is **not relevant**.

- If the **conditioning event**, or signal, is **independent from the event we are interested in**, then it does not help predict it
- In this case, the signal, is not **informative**



What is the joint probability of rolling two 6's?



Joint probability with independent events: rolling two dices

EVENT 1

probability of rolling 6 in
the first roll = $1/6$

EVENT 2

probability of rolling 6 in
the second roll = $1/6$

JOINT PROBABILITY OF EVENT 1 AND 2

probability of (EVENT 1)
X probability of (EVENT
2) = $1/6 \times 1/6$

Joint probability: $P(X,Y) = P(X|Y) \cdot P(Y)$

But, if X and Y are independent, $P(X|Y) = P(X)$

So, in case of independence:

Joint probability = $P(X|Y) \cdot P(Y) = P(X) \cdot P(Y)$

In this case, X = probability of rolling 6 in the first round
Y = probability of rolling 6 in the second round



		Area of origin			
		Italy	Asia	Others	MARGINAL
# letters in last name (X)	2	0	2	0	2
	3	1	5	0	6
	4	3	1	0	4
	5	9	2	2	13
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	9	7	0	2	9
	10	4	0	0	4
	11	1	0	0	1
	14	0	0	2	2
	23	0	0	1	1
MARGINAL		63	12	18	93

Independence is a matter of degree!

$$P(X=5 \mid \text{Italy}) = 9/63 = \mathbf{14.3\%}$$

$$P(X=5) = 13/93 = \mathbf{14.0\%}$$

Although the two events (five letters and being Italian) are not independent, the conditional probability (five letters when you are Italian) is very close to the marginal probability of having five 5 letters...

So, being Italian is not a very informative signal.



Area of origin					
		Italy	Asia	Others	MARGINAL
# letters in last name (X)	2	0	2	0	2
	3	1	5	0	6
	4	3	1	0	4
	5	9	2	2	13
	6	8	2	4	14
	7	21	0	5	26
	8	9	0	2	11
	9	7	0	2	9
	10	4	0	0	4
	11	1	0	0	1
	14	0	0	2	2
	23	0	0	1	1
MARGINAL		63	12	18	93

What about whether being Asian or Italian predicts whether your last name has 3 letters?

$$P(X=3 \mid \text{Asia}) = 5/12 = 41.7\%$$

$$P(X=3) = 6/93 = 6.4\%$$

$$P(X=3 \mid \text{Italy}) = 1/63 = 1.6\%$$

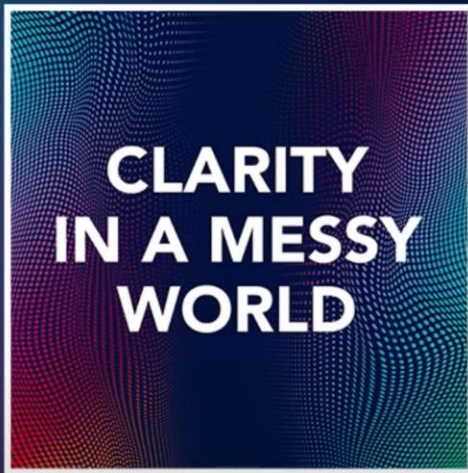
$$P(X=3) = 6/93 = 6.4\%$$

Which one is more informative?

Remember: prediction can go either ways



Optional: podcast episode on using scientific thinking in business decision making



Episode #5
Why Entrepreneurs
Should Act like Scientists
with Alfonso Gambardella



5.0

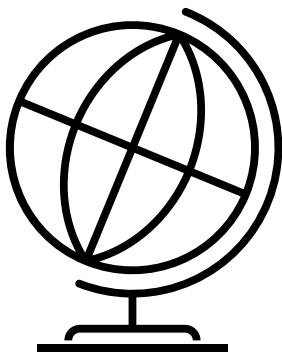
PREDICTION OF EVENTS



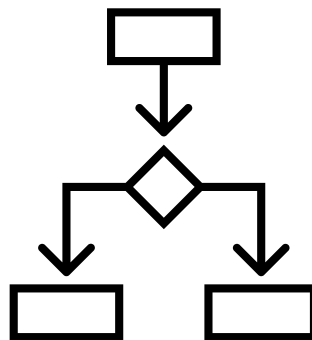
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Why is probability relevant to management?

PREDICT EVENTS



DECIDE ACTIONS



Why is the new robot failing?

Your company has recently bought a new robot for the factory, which has shown signs of malfunction. The robot starts regularly but after a few hours, on certain days, it comes to a halt all of a sudden.

After thinking about why this may be the case, you hypothesize that the robot fails **when the temperature in the factory is low.**

So, you ask your engineers to have a look at the factory records and collect the temperatures of the days the robot failed.



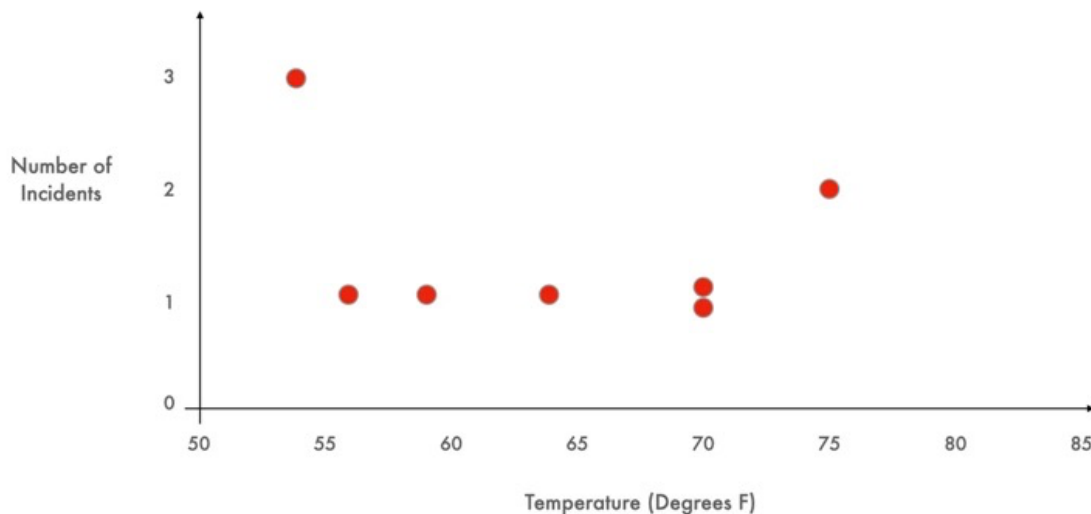
Poll time!

Are the failures of the robot due to low temperature?

A) Yes

B) No

The engineers came back with this graph, which tells you the number of accidents at different recorded temperatures.



50 degrees F = 10 degrees C

75 degrees F = 23.9 degrees C



Think again

Your theory is an if-then statement.

If the temperature in the factory is low, then the robot fails.

How do you test this theory?

By looking at **all the days** in which the temperature is low, and then checking whether the robot fails on those days.

It's like the theory: "If mum is happy, she gives me chocolate". You want her to give you chocolate **every time** she is happy.



You need more data

The engineers gave you only the temperatures of **the days when the robot failed**.

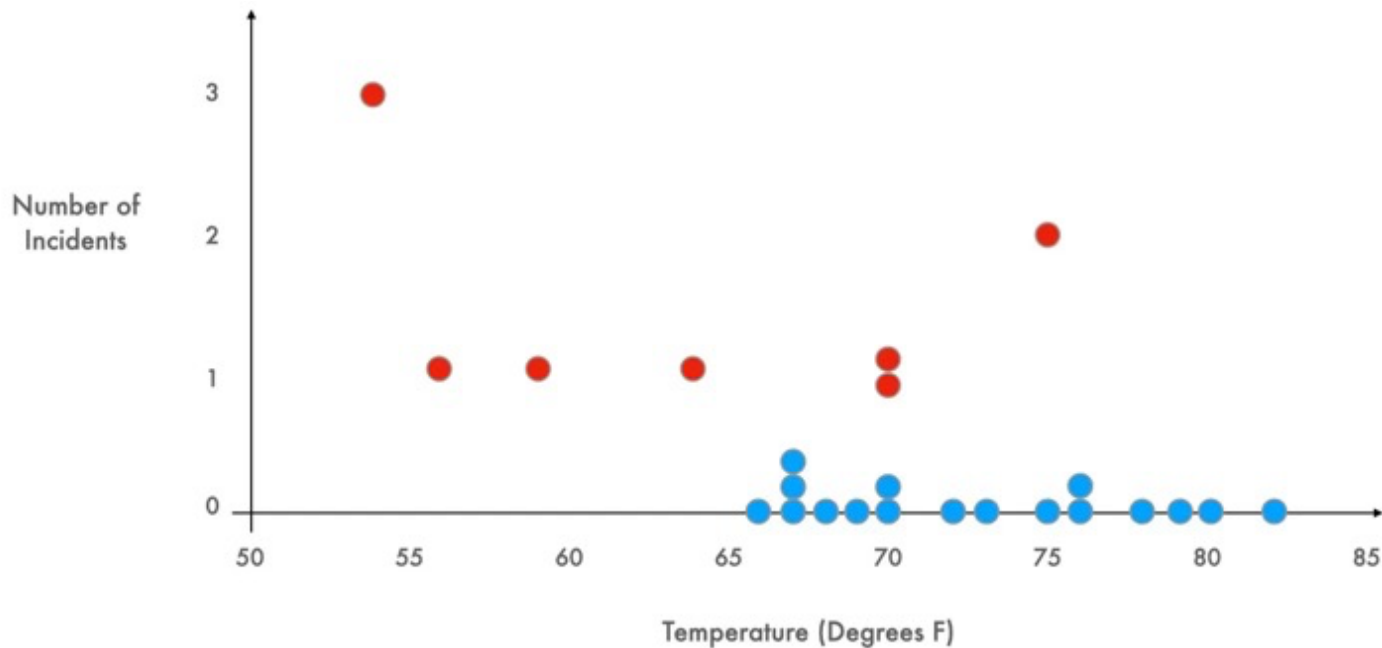
What about the **days without incidents**? Maybe the temperature was low on those days too and nothing happened...

You tell the engineers.

They come up with a new graph with the temperatures of all the days in which the robot operated.



The complete graph



What's your conclusion now?

Table version

	Days with accidents	Days without accidents	Total
Low temp (< 65° F)	6	0	6
High temp (≥ 65° F)	4	17	21
Total	10	17	27

The first graph showed you only the first column (days with incidents). You could calculate:

$$P(\text{Low temperature} \mid \text{accident}) = 6/10 = 60\%$$

But, to test your theory, what you need is the full table to test this probability:

$$P(\text{accident} \mid \text{Low temperature}) = 6/6 = 100\%$$



It wasn't actually a robot



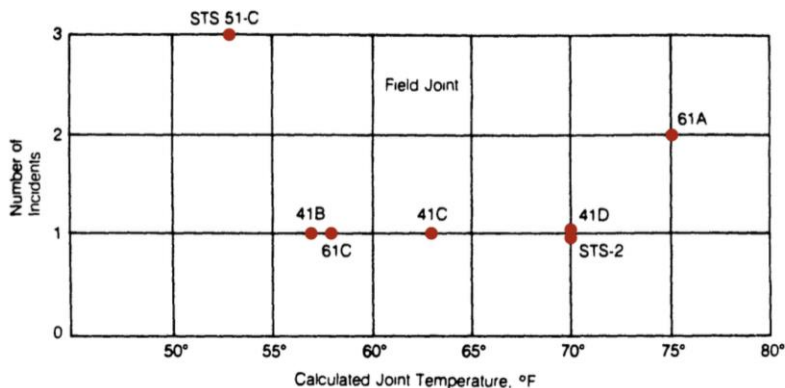
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Archival: Space Shuttle Challenger Disaster | NBC Nightly News

<https://www.youtube.com/watch?v=yibNEcn-4yQ>

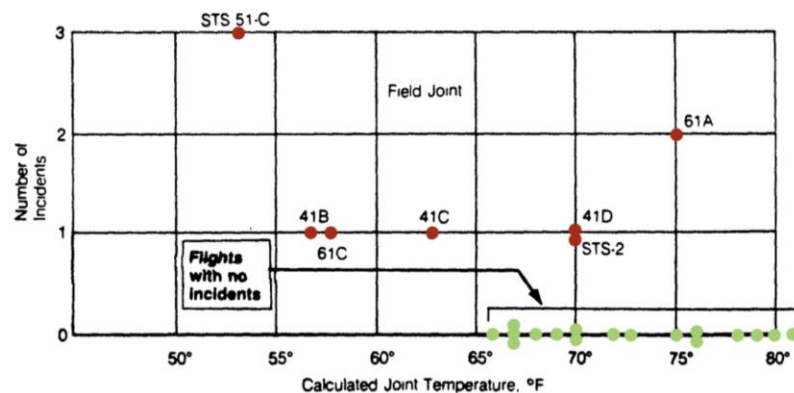
It wasn't actually a robot

Number of O-Ring incidents vs. Joint Temperature
(Incidents when O-Rings failed)



Source: Report of the Presidential Commission on the Space Shuttle Challenger Accident, 6 June 1986, Volume 1, Page 145, ([link](#)) Color added.

Number of O-Ring incidents vs. Joint Temperature
(failures AND successes)



Source: Report of the Presidential Commission on the Space Shuttle Challenger Accident, 6 June 1986, Volume 1, Page 145, ([link](#)) Color added.



Definition of **theory**

- A series of logical steps linking antecedents to consequences
- If P then Q, if Q then R, if R then S, ..., if V then Z ... You eventually show that if P then Z



Why are theories important?

- Theories tell you what data / factors to examine
- The factors that affect your outcome improve your ability to make an accurate prediction
- The data that you find provide signals for your conditional probabilities, i.e., for improving your predictions

In practice:

1. You theorize “if A, then B”
2. You collect all the cases with A, and then you measure B in them
3. You calculate $P(B|A)$
4. If needed, you repeat with a theory about A, and so on



We need to pay attention to the data we collect (representative, large sample)

The Literary Digest

NEW YORK

OCTOBER 31, 1936

Topics of the day

LANDON, 1,293,669; ROOSEVELT, 972,897

Final Returns in The Digest's Poll of Ten Million Voters

Well, the great battle of the ballots in the Poll of ten million voters, scattered throughout the forty-eight States of the

Republican National Committee purchased THE LITERARY DIGEST? And all types and varieties, including: "Have the Jews purchased

returned and let the people of the Nation draw their conclusions as to our accuracy. So far, we have been right in every Poll. Will we be right in the current Poll? That, as Mrs. Roosevelt said concerning the President's reelection, is in the 'lap of the gods.'

"We never make any claims before election but we respectfully refer you to the opinion of one of the most quoted citizens



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PRESIDENT ROOSEVELT REELECTED

A NEWSPAPER
FOR A
CONSTRUCTIVE
POLICY

ATLANTA DAILY WORLD
"The Nation's Only Colored Daily Newspaper"

CITY
EDITION

Lauds Gandhi In Stirring Chapel Talk

Roosevelt Leads Landon In Landslide As Polls Show Endorsement Of New Deal

Roosevelt Haring Returns

Re-elected President Of United States

Chest Drive Plans Begin For Coming Campaign; High Goal

Landon Listening to Returns

Officers Meet In Chase Of 'Runners'

Prof G. W. Eagleson Heard In Talk On Abnormal Behavior

AME's Open Eventful Session Of North Georgia Conference

Janitors' Group YWCA To Stage Open House Tonight At Library

Today

How to predict events

You have a target event

You want to predict whether it will occur

Example: are there at least 10 supporters of Inter Milan in this class?



Step 1

We need a theory!



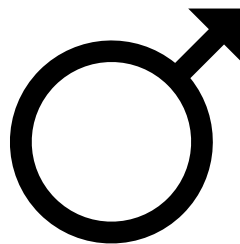
Step 1

What's
your
theory?

What theory do you have to evaluate whether or not there are at least 10 supporters of Inter Milan in the class?

What factors would you use to build your theory?

Maybe you look around and check how many male students there are in the class. Or, how many are from Milan.



Step 1

What's your theory?

Based on your theory, you envision a certain probability (e.g. 60%) that there can be at least 10 supporters of Inter Milan in the class.

This is called **prior probability**.

Prior probability is **the predicted probability of an event before data is collected**. It is the best assessment of the probability of an outcome based on the current knowledge

Write down your prior probability!



Step 2

Revise your theory

Now that you have a prior probability you decide whether to continue with this probability or to **look for more information/signals**.

Signals are costly and you usually cannot collect too many of them.

At least two ways to collect signals:

1. **Ask others** (experts)
2. **Collect data** (data analysis)

How do you apply them to your research on Inter Milan supporters?



Step 2

Revise your theory:

- Ask others

Try to improve your prior probability by **asking the classmates** sitting close to you what they think the probability of having at least 10 students supporting Inter Milan is.



Step 2

Revise your theory:

- Ask others

- They say 80% ... In this case you may update your probability to a higher level than your prior of 60% (e.g. to 70%)
- They say 40% ... in which case you update your probability to a lower level than your prior of 60% (e.g. to 50%)

The new probability is a **posterior probability**.

Posterior probability is the revised probability of an event occurring after taking new information into account.

Write down your posterior probability!



Step 2

Revise your theory:

- Collect data



Let's **collect data** now.

Take a sample of 20 students in the class and check how many of them are supporters of Inter Milan.

- If they are at least 2 (10%), update your posterior probability **upward**
- If not, update your posterior probability **downward**

**Write down your NEW
posterior probability!**

Predict events

STEP 1 – THEORY

You look at the class (100 students) with a theory in mind.

You set a **prior probability**.

STEP 2 – LOOK FOR MORE INFORMATION

Then you ask friends (experts and collect data).

You update you prior probability (multiple times), generating **posterior probabilities**

STEP 3 – DEFINE YOUR EXPECTATIONS

You decide when to stop collecting data/signals and to finalize **your probability**, that is, a posterior probability that you cannot improve further



Technically

$$P(Y) = 60\%$$

PRIOR PROBABILITY

Y = at least 10
students in the class
support Inter Milan

**$P(Y|S)$ = PROBABILITY OF
EVENT Y GIVEN THE
OBSERVED SIGNAL S**

S: what the data or your friends
told you

— If $S = 80\% \rightarrow P(Y|S=80\%) = 70\%$

— If $S = 40\% \rightarrow P(Y|S=40\%) = 50\%$



Technically

There could be a series of priors, signals, posteriors, in which each posterior becomes the prior before the next signal

$$P(Y) \xrightarrow{s_1} P(Y | s_1) \xrightarrow{s_2} P(Y | s_1, s_2) \xrightarrow{s_3} P(Y | s_1, s_2, s_3)$$

For example:

- $P(Y) = 60\%$,
- Data tell you $S_1 = 40\%$,
- You update $P(Y | S_1 = 40\%) = 50\%$
- New data collection tells you $S_2 = 45\%$
- You update $P(Y | S_1 = 40\%, S_2 = 45\%) = 48\%$
- And so on....





**So... how many supporters
of Inter Milan in this class?**



6.0

MAKING DECISIONS



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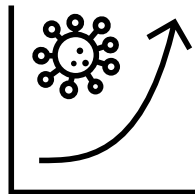
From predicting events to making decisions and estimating payoffs



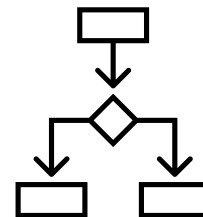
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WHY DO WE NEED TO PREDICT EVENTS?

An event is
associated with
the outcome of an
action



You want to
decide whether or
not to take that
action



From predicting events to making decisions and estimating payoffs



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WHY DO WE NEED TO PREDICT EVENTS?

Example

Pay 1,000 € upfront, and you receive 100 € per student who support Inter Milan in this class (100 students). Therefore:

- if at least 10 students support Inter you break-even (total cost = 1,000€; total income = $100€ \times 10 \text{ students} = 1,000€$)
- if there are less than 10 supporters, you lose money
- if there are more than 10 supporters, you gain money

Should you pay 1,000€
upfront? How to approach
the decision?



You are facing this situation

Actions \ Scenarios	Scenarios	
	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	PROFIT	LOSS
Not pay 1,000 upfront	0	0

- **Actions** = actions you are going to take
- **Scenarios** = states of the world you are going to face



You are facing this situation

Actions \ Scenarios	Scenarios	
	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	PROFIT	LOSS
Not pay 1,000 upfront	0	0

- **Actions** = actions you are going to take
- **Scenarios** = states of the world you are going to face

This is the set of options
that is available to you.
You decide on this.



You are facing this situation

Actions \ Scenarios	Scenarios	
	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	PROFIT	LOSS
Not pay 1,000 upfront	0	0

- **Actions** = actions you are going to take
- **Scenarios** = states of the world you are going to face

This is what could happen. You do not control it. This is what you PREDICT.



You are facing this situation

Actions \ Scenarios	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	PROFIT	LOSS
Not pay 1,000 upfront	0	0

- **Actions** = actions you are going to take
- **Scenarios** = states of the world you are going to face

These are the outcomes of each action-scenario. This is what you are assessing



You are facing this situation

Actions \ Scenarios	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	PROFIT	LOSS
Not pay 1,000 upfront	Neither profit or loss	Neither profit or loss

How to proceed?

Intuitively, the HIGHER the probability that at least 10 students support Inter Milan, the more likely it is that you get a PROFIT.

PREDICT = estimate the probability of the different outcomes



How do you decide?

STEP 1 – THEORY

You look at the class (100 students) with a theory in mind.

You set a **prior probability**.

STEP 2 – ELICIT SIGNALS

Then you ask friends (experts and collect data).

You update you prior probability (multiple times), generating posterior probabilities

STEP 3 – DEFINE YOUR PROBABILITY

You decide when to stop collecting signals and to finalize your **probability**, that is, a posterior probability that you cannot improve further

You have already worked on probability estimation.

You have a final posterior probability.

But assume you do not know the real value yet (whether there are at least 10 supporters)

Assume your probability is 52%. Do you pay 1,000 upfront?

And if it is 55%?

And if it is 60%?

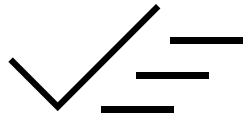
You need a criterion



Establish a criterion

APPROACH 1

Think of a
threshold
probability



APPROACH 2

Assign values
(profits and losses)



1. Threshold

You think of a **threshold** P^* such that:

- if your expected payoff $P(Y|S) \geq P^*$, then you go ahead with the action
- if your expected payoff $P(Y|S) < P^*$, then you don't

IMPORTANT

Decision-makers should select the threshold **before** they estimate their expected payoff or see the data.

Doing it after often generates confirmatory biases (you choose the threshold you want) leading to bad decisions.



I recently went clothes shopping in Brooklyn with my friend Emma. Showing off a pretty dress, she tugs at the price tag on the back. “Hey, what does this say?” she asks me. “If it’s less than 80 bucks, I’ll buy it.” Now that’s some decision intelligence! Instead of first seeing the price and then talking herself into a decision she’s already made, she uses the data to drive it. With a well-practiced reflex, she weighs how much she likes the dress and her budget, then sets the decision boundary ...

<https://hackernoon.com/data-inspired-5c78db3999b2>

Cassie Kazyrkov | Chied Data Scientist at Google

2. Assign values

Given the probability that you have estimated, you assign values to profits and losses in the action-scenario matrix.

Two approaches.

- **Value model** (you calculate monetary values of profits and losses that you could make in the different scenarios)
- **Utility model** (you attribute «utilities» that represent how much you enjoy a positive outcome or how much you suffer a loss)



Assign values: Value model

Actions \ Scenarios	Scenarios	
	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	+500€	-500€
Not pay 1,000 upfront	0	0

You estimate that:

- if there are at least 10 students supporting Inter, on average there are 15 students supporting Inter → revenue – cost = $15 \times 100\text{€} - 1000\text{€} = 500\text{€}$
- if there are fewer than 10 students supporting Inter, on average there are 5 students supporting Inter → revenue – cost = $5 \times 100\text{€} - 1000\text{€} = -500\text{€}$



Compute the expected value. Is it greater than 0?



Assign values: Value model

Actions \ Scenarios	Scenarios	
	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	+500€	-500€
Not pay 1,000 upfront	0	0

Expected value (Pay 1,000) = $500 * P(Y|S) - 500 * (1 - P(Y|S))$

Expected value (Not pay 1,000) = $0 * P(Y|S) + 0 * (1 - P(Y|S)) = 0 + 0 = 0$

Example: $P(Y|S) = 55\%$

Expected value (Pay 1,000) = $(500 * 0.55) - (500 * (1 - 0.55)) =$
 $= (500 * 0.55) - (500 * 0.45) =$
 $= 275 - 225 = 50$



CRITERION: You choose the action with the greater expected value.

Assign values: Value model

The value model derives a threshold probability based on the **expected outcomes**.

Threshold in the value model: the minimum probability required to make the expected value ≥ 0 .

The Expected value is $500 * P(Y|S) - 500 * (1 - P(Y|S))$

We want: Expected value ≥ 0 .

It is then easy to see that this inequality is true if $P(Y|S) \geq \frac{1}{2}$, which will then be your threshold ($P^* = 50\%$)



Assign values: Utility model

Utilities: how much you enjoy a positive outcome or how much you suffer from experiencing a loss

This depends on:

- Monetary values (money is useful...)
- Other aspects of the decision

E.g. you may be very worried about losing money (risk aversion) and the loss of 500 € has a disutility of 400

Or you are very excited if you run a successful project, and turn the profit of 500 € has an utility of 800

Then you calculate the **Expected utility** of the actions (same as expected value, but with utilities in place of monetary values).



Assign values: Utility model

Actions \ Scenarios	Scenarios	
	At least 10 Inter supporters	Less than 10 Inter supporters
Pay 1,000 upfront	800	-400
Not pay 1,000 upfront	0	0

For instance, you set utility equal to 800 if the project succeeds; -400 if it does not.

The expected utility is $800 \cdot P(Y|S) - 400 \cdot (1 - P(Y|S))$.

Threshold: Expected utility ≥ 0 when $P(Y|S) > 1/3$ or $P^* = 33\%$

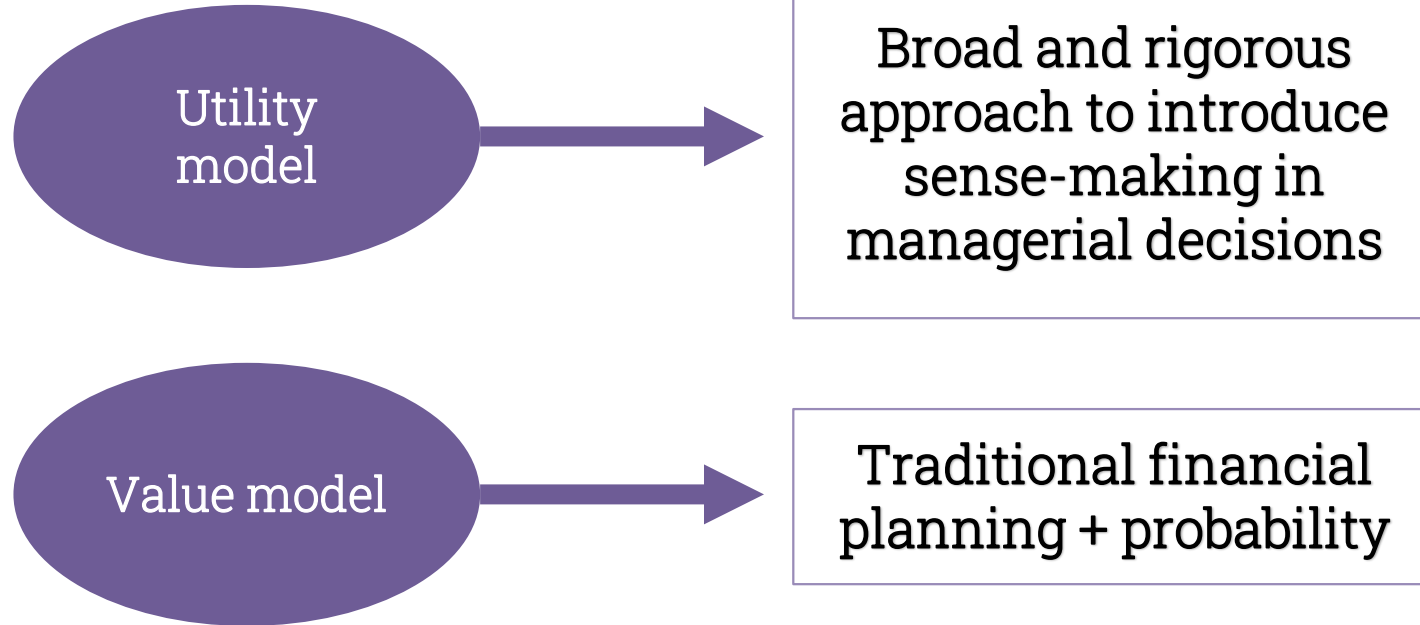


Value model vs Utility model

1. The utility model or the value model are formally equivalent. You can see the value model as a special case of the utility model in which your utility depends only on monetary value.
2. The difference is that you base utility on what is important to you



Value model vs Utility model



Wrap up: A framework for managerial decision-making

DEFINE ACTIONS AND SCENARIOS

What are the possible courses of action? What can you control? What you cannot?

DEFINE A MAP OF ACTION-SCENARIO PAIRS

What are the possible combinations?
Assign the payoffs derived from pair.

ESTIMATE THE PROBABILITY

Have a theory, estimate your priors, and decide whether to revise them

MAKE A DECISION

Based either on a predefined threshold or on monetary values/utility



The framework can accommodate complex decisions

Your scenario-action pairs can also have more actions and scenarios, such as the following:

Actions \ Scenarios				
	High demand	Medium demand	Low demand	Very low demand
Produce green hats	profit	profit	profit	loss
Produce red hats	profit	profit	loss	loss
Produce blue hats	profit	profit	loss	loss
Do not produce hats	0	0	0	0

In this case you still estimate the probability of the different scenarios, find the highest expected value given these probabilities, and take the action associated with the highest expected value (which could be the default action)



7.0

APPLICATION: SOUTHWEST AIRLINES



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Should Southwest introduce fees?

We are in 2016.

- Analysts are pressuring Southwest to introduce fees to increase revenue
- Southwest's CEO is worried that fees could undermine customer trust

How would you apply our framework to this decision?

Why Wall Street Isn't Happy With Southwest's 43 Straight Years of Profits

skift.com/2016/08/16/why-wall-street-isnt-happy-with-southwests-43-straight-years-of-profits

August 16, 2016



Southwest has made money for more than 40 years. But that consistency does not satisfy many Wall Street analysts, who keep pushing the airline to increase profits.

— Brian Sumers



1. Actions

General recommendation: let's keep the model **as simple as possible**, e.g.
minimum number of actions, scenarios,
numbers that are easy to calculate, basic
theories, etc.

There is always time to complicate it
later if necessary!

**WHAT ACTIONS ARE AVAILABLE TO
SOUTHWEST AIRLINES?**



1. Actions

2. Scenarios

Which scenarios are relevant to the actions we are considering?

- They must be important for our decision
- They should be out of our control (once we take the action)
- We have to be able to estimate the probability that they happen
- They should produce payoffs (values/utilities) (depending on our actions)

WHICH SCENARIOS SHOULD YOU CONSIDER?



1. Actions
2. Scenarios
3. **Decision model**

Let's choose a decision making framework. Three options:

- Threshold
- Value Model
- Utility Model

It's important that you choose a model you can actually use, given the information that you have.

**WHICH DECISION MAKING FRAMEWORK
DO YOU WANT TO USE?**



1. Actions
2. Scenarios
3. Decision model
4. **Utilities**

Actions \ Scenarios	Scenarios	
	Customer trust not impacted	Customer trust damaged
Introduce fees	?	?
Not introduce fees	?	?

The utility derived should be linked to the strategy that Southwest Airlines is pursuing.

WHAT UTILITIES DO YOU WANT TO INPUT?



1. Actions
2. Scenarios
3. Decision model
4. **Utilities**

Actions \ Scenarios	Scenarios	
	Customer trust not impacted	Customer trust damaged
Introduce fees	100	-300
Not introduce fees	0	0

We can assign utilities to reflect the serious damage on Southwest that results from implementing a strategy that damages customer trust



1. Actions
2. Scenarios
3. Decision model
4. Utilities
5. **Threshold**

Actions \ Scenarios	Scenarios	
	Customer trust not impacted	Customer trust damaged
Introduce fees	100	-300
Not introduce fees	0	0

We can now calculate how probable we want the event "customer trust not impacted" to be, in order to introduce fees.

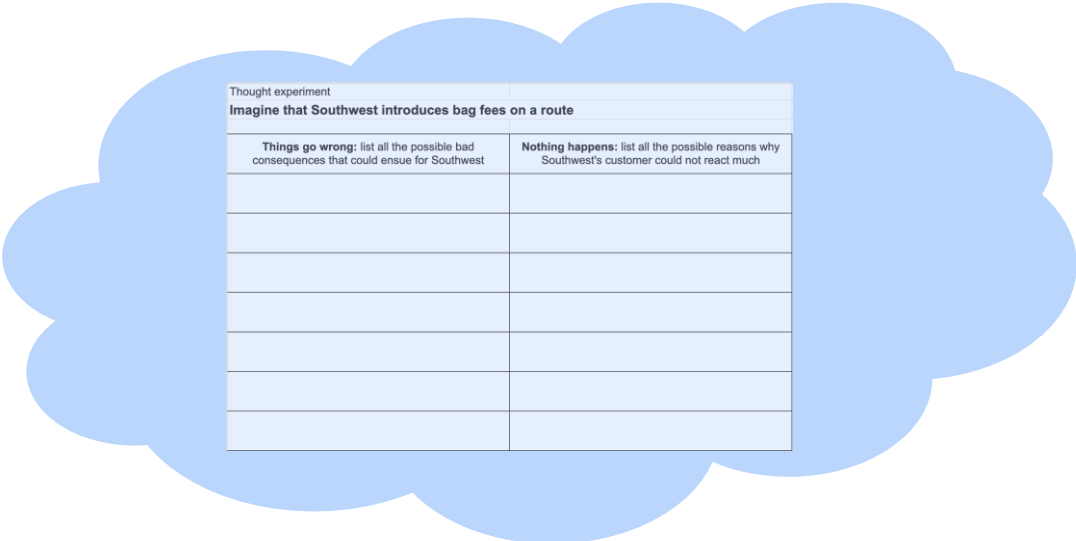
WHAT IS THE THRESHOLD?



1. Actions
2. Scenarios
3. Decision model
4. Utilities
5. Threshold
6. **Estimate probability**

The threshold is 75%. We can now ask ourselves:

WHAT IS THE PROBABILITY THAT CUSTOMER TRUST IS NOT IMPACTED BY THE INCREASE IN FEES?



Thought experiment	
Imagine that Southwest introduces bag fees on a route	
Things go wrong: list all the possible bad consequences that could ensue for Southwest	Nothing happens: list all the possible reasons why Southwest's customer could not react much



1. Actions
 2. Scenarios
 3. Decision model
 4. Utilities
 5. Threshold
 6. Estimate probability
- **Have a theory**

Have a theory: What are the antecedents of customer trust, for an airline company?

Theory: If an airline company is ..., or does ..., then customers trust it.

For example, research shows that trust depends a lot on keeping promises. Is this relevant in this case?

WHAT IS YOUR PRIOR THAT CUSTOMER TRUST IS NOT IMPACTED BY THE INCREASE IN FEES?



1. Actions
2. Scenarios
3. Decision model
4. Utilities
5. Threshold
6. Estimate probability
 - Have a theory
 - **Ask experts**

Ask experts: Who could you speak to, or what data can you look for, to improve your prior?

- Consultants
- Equity analysts
- Experienced Southwest's managers
- What the other airline companies have already done (WestJet, JetBlue)
- Loyal customers (surveys, focus groups)
-

DO YOU WANT TO UPDATE YOUR PRIOR?



1. Actions
 2. Scenarios
 3. Decision model
 4. Utilities
 5. Threshold
 6. Estimate probability
- Have a theory
 - Ask experts
 - **Collect data**

Collect data: You could run an experiment on a route or few routes, as suggested by the J.P. Morgan's analyst.

Remember: signals can be costly.

Which is the cost of the signal, in this case?

AFTER THE EXPERIMENT (IF YOU RUN IT), YOU ARRIVE AT YOUR FINAL ESTIMATE

- You make your decision based on whether the final estimate is larger or smaller than the threshold



What did Southwest do?

It has never
changed its fee
policy.

It has never ran
an experiment
either.



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Southwest® Boarding Process

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How does the boarding process work?

You will be assigned a boarding group (A, B, or C) and position (1-60+) upon check in. Your unique group and position combination (for example: A35) will be displayed on your boarding pass and represents a reserved spot in the boarding group at the gate. Numbered posts in each of our gate areas indicate where to line up. When your boarding group is called, find your designated place in line and board the aircraft in numerical order with your boarding group.

Do I have an assigned seat?

Southwest-operated flights have open seating. Once onboard, simply choose any available seat and stow your carryon items in the overhead bin or under the seat in front of you.

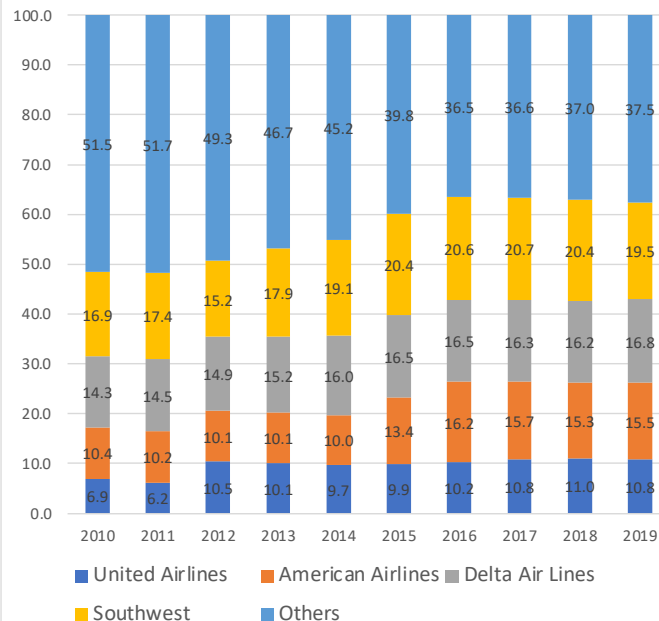


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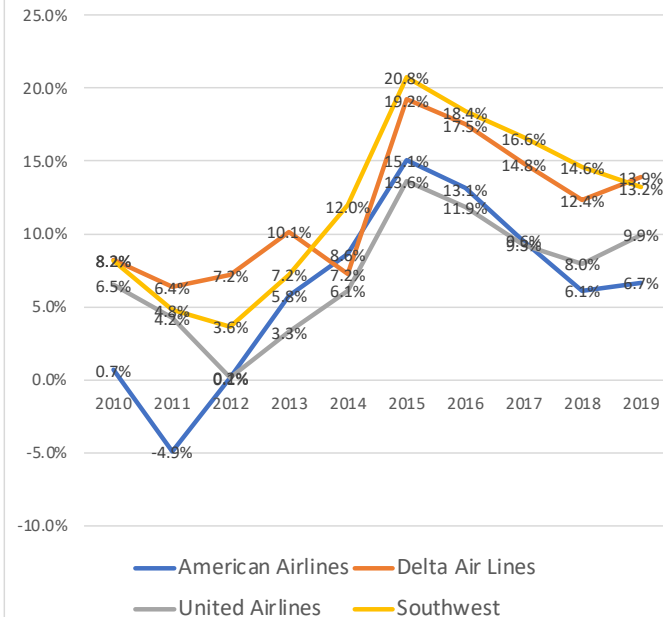
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How did Southwest do?

US domestic flights % market share (2010-2019)



Operating margin (2010-2019)



THANK YOU